

Hall Effect Sensors: Big Things in Small Packages

Hall effect sensors were first used as microwave power sensors. These now pervade everything from automobiles to computers to machine tools. Take a peek into the different types of Hall effect sensors



T.K. HAREENDRAN

Since its introduction several years ago, the Hall effect sensor has captured the imagination of design engineers. Being solidstate, it is a more reliable approach than electromechanical counterparts. Since the Hall effect sensor has no moving parts, low-level loads last well beyond billions of operations. There are many applications of Hall effect sensors. However, selection of the proper type is critical for any application.

What's a Hall effect sensor

Sensors can be classified based on the physical effects that they can respond to. Hall effect sensors are unique in that their principle of operation is based on mixed effects.

When a current-carrying conductor is placed into a magnetic field, a voltage is generated perpendicular to both the current and the field.

This principle is known as the Hall effect. So Hall effect sensors depend on the electric current as well as the magnetic field.

Typically, a Hall effect sensor has three wires or terminals: one for ground, one for supply (or reference) voltage and one for the output signal. The supply voltage is necessary to create the switching effect that takes place inside the

sensor.

In practice, the basic magnetic field sensor—Hall element—is constructed from a thin sheet of conductive material with output connections perpendicular to the direction of current flow. When subjected to a magnetic field, it responds with an output voltage proportional to the magnetic field strength. The voltage output is in micro-volts (μV) range and requires signal conditioning electronics to achieve useful voltage levels.

The additional electronics needed is a differential amplifier and a temperature compensation circuit. Voltage regulation is also required when operating from an unregulated supply. When the Hall element is combined with this associated

electronics, it forms a ready-to-use Hall effect sensor, which is an integrated circuit chip that contains a Hall element and the signal conditioning electronics.

According to Honeywell's *Hall Effect Sensing and Application*, although Hall effect sensor is a magnetic field sensor, it can be used as the primary component in many other types of sensing devices. If the quantity to be sensed incorpo-

Classification of Sensors Based on Their Sensing Principle

Sensing principle	Type of sensor
Mechanical motion	Tilt sensor, piezo-resistive pressure sensor
Thermal energy	Thermocouple, thermistor
Magnetic field	Hall effect sensor, reed switch
Optical energy	Photo-diode, photo-resistor
Electrical current	Hall effect sensor

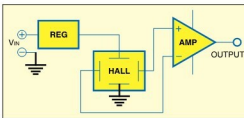


Fig. 1: Basic analogue-output Hall effect sensor

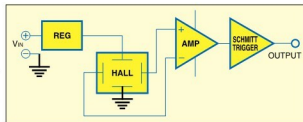


Fig. 2: Basic digital-output Hall effect sensor